# Peer Learning Activity: An Enterprise Distributed Computing Application

## Motivation:

Any bachelor-level coursework usually involves learning to be an effective user and an effective developer/builder. For example, consider your CSCE A365 Computer Networks Course. You learned how to use the Networks and the Internet effectively. You also learned how to build networks, how to build *an Internet* (such as through those BGP Peering discussions), and how to develop novel protocols. Consider your Operating Systems class. You learned how to use major operating systems (Windows and UNIX) and also how to build those operating systems (the research aspects). A Machine Learning class would also teach you how to use and develop ML algorithms.

This course is no different. You are learning how to develop Distributed Computing algorithms and applications. At the same time, you also learn how to effectively use existing Distributed Computing software, especially those that are used in the industry.

## Task:

As part of this task, you will form a group and work on an enterprise Distributed Computing platform. Several of those will be introduced to you in class on Aug 27th. Further classes will continue the conversation with more frameworks. Students can also find additional frameworks on their own.

Your task is to, as a team, effectively use the chosen product for one or more distributed computing task(s) by **carefully going through its existing examples and tutorials.** You may use Assignment 1 as a stepping stone towards this. However, there is a difference in the goals. Assignment 1 aims at solving a problem that you define. On the other hand, this Peer Learning Activity expects you to get familiarized in a chosen platform, get familiarized as a user, and be comfortable to present it to an audience through existing and expanded use cases and tutorials. Moreover, this is teamwork. So, you will still need to come to a consensus on what framework you are using with your teammates. Also, we should avoid everyone presenting the same framework. It is quite unlikely, but not impossible. Discuss with the instructor once you have formed a team and chosen a framework. Apache Software Foundation (ASF https://projects.apache.org/) has many relevant distributed computing projects that you can pick from. Google Summer of Code (GSoC https://summerofcode.withgoogle.com/programs/2025/organizations) also has many organizations championing distributed computing. But be aware that not all the ASF and GSoC projects are relevant. So, you should choose an appropriate one and confirm with the instructor before investing time on one.

Another potential framework is Infinispan. This is one of the **In-Memory Data Grid** platforms. However, this is mentioned only as an example. Please do not go with Infinispan just because it is mentioned as an example. Students are free to choose a product/project that works best for them. You could try several and find one that fits your interest (as a team). You could also use multiple products in conjunction, although that is not expected. If you decide to use multiple frameworks, make sure you still give sufficient attention to (at least) one project. We don't want a collection of platforms vaguely discussed.

Please pay attention. Some of these frameworks are truly free and open source (FOSS). All ASF and GSoC projects are truly FOSS. Others require payment for a license. Many fall somewhere in between, in a freemium model (free to try, but must pay for a production deployment). Find one that does not require you to pay for evaluation. Consult the instructor with your questions, if you are unsure.

## What we learn from this peer learning activity

This is a peer-learning activity. The goal is to promote peer learning, and that is achieved through two steps. First, teams of variable sizes are formed, and students work in those teams. Second, a demo (and optionally a presentation) by each team on the date as set on the Syllabus takes the form of Peer Learning Activity Discussions led by students. During the demo, students must have installed and configured the chosen framework and present its capabilities through a demo, in a tutorial style.

Although the activity is a team-based endeavor, it is graded individually. Therefore, everyone must contribute to their team's implementation and work, as well as the demo/presentation. The small class size allows flexibility in the size of the teams, although 2 – 3 member teams are encouraged. There is also flexibility in the presentation aspect. If a student prefers to write down their observations as a report rather than contributing to their group's demo/presentation, that is fine as well. However, please inform the instructor if that is your choice. If you are not present during the "Peer Learning Activity Discussions" on the indicated date AND if you fail to submit a report explaining your achievements as part of this activity, you may be awarded a zero point (since there is no way to evaluate your progress!).

## Rubric

The assignment is evaluated across two different aspects. Each aspect is judged on a scale of 0 – 4, from 0 being "Ungradable" to 4 being "Excellent.

- **Excellent** (4): The work is either perfect or has negligible flaws.
- **Good** (3): The work is of good quality. Only minor revisions would be needed for the work to be Excellent.
- **Satisfactory** (2): While the work has some room for improvement, the student has put in a good-faith effort to complete all the work and demonstrated sufficient mastery of the material. A few revisions would be needed for the work to be considered Good.

- **Needs Improvement** (1): The student has put in a good-faith effort to complete the work, but revealed a lack of mastery in the material that can be addressed via concrete feedback. The work could become Satisfactory or better with some major revisions.
- **Ungradable** (0): The student did not make a good-faith effort to complete the work. This includes not submitting the (relevant part of) work at all, but also situations like submitting only placeholder code or code segments merely taken from the textbook.

**The grading aspects:**

1. **Completeness** in Distributed Computing aspects: As demonstrated by the demo (and optionally a presentation) or a detailed Individual report, the student has demonstrated their ability to use the chosen product effectively to resolve a distributed computing challenge. If it is a report, plots/figures showing the numbers are encouraged, rather than simple screenshots of the product running.
2. Ability to explain the findings to **educate** others in Distributed Computing Aspects: Through either the interactive hands-on demo (probably supported by a Powerpoint presentation or using the whiteboard) or the Individual report, the student establishes their strength in educating others on how to use the chosen platform effectively for distributed computing tasks, such as message passing, scheduling, replication, distributed execution, and fault tolerance.

The remaining two points come from participation in the Peer Learning Activity Discussions, which are graded 0, 1, or 2. (Partial points such as 0.5 and 1.5 are possible based on situations).

**Absent** (0): The student was not present during the discussions.

**Passive** (1): The student was present but largely passive and did not contribute to the conversations or discussions.

**Active** (2): The student was active throughout the Peer Learning Discussion. They asked questions, clarified points, and contributed positively to the peer learning aspect of other teams' projects.